Functions of Random Variables

Finding the Probability Distribution of a function of Random Variables

Example

Let $x \sim N(\mu, \sigma^2)$.

and $Y = a + bx$.

Find the pdf of Y.

Solution

$$x \sim N(\mu, \sigma^2)$$

$$E(x) = \mu, \quad V(x) = \sigma^2$$

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-1}{2}(x-\mu)^2}, \quad -\infty < x < \infty$$

To find the p.d.f. of Y via a method of moment.

$$Y = a + bx.$$

$$E(Y) = E(a + bx)$$

$$= a + bE(x)$$

$$= a + b\mu.$$

$$\therefore E(Y) = a + b\mu$$

Also $V(Y) = V(a + bx)$

$$= V(a) + V(bx)$$

$$= b^2 V(x)$$

$$= b^2 \sigma^2.$$

$$\therefore V(Y) = b^2\sigma^2.$$

Thus $Y \sim N(a+b\mu, b^2\sigma^2)$.

The pdf of the random variable Y is

$$f(y|a+b\mu, b^2\sigma^2) = \frac{1}{\sqrt{2\pi}\,b\sigma} e^{-\frac{1}{b^2\sigma^2}(y-(a+b\mu))}.$$

Example 2

Let X be a Bernoulli(θ) random variable and suppose $Y = X_1 + X_2 + \cdots + X_n$; Find the pdf of Y.

Soln

$X \sim B(\theta)$.

$P_X(x) = P[X=x] = \theta^x (1-\theta)^{1-x}$

$E(X) = \theta$ and $V(X) = \theta(1-\theta)$.

Then $Y = \sum_{i=1}^{n} x_i$.

$$E(Y) = E\left[\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} [E[x_i]] = \sum_{i=1}^{n} \theta_i = n\theta.$$

Also

$$V(Y) = V\left[\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} [V[x_i]] = \sum_{i=1}^{n} [\theta(1-\theta)] = n\theta(1-\theta).$$

$E(Y)$ is the mean of a Binomial random Variable with parameter n and θ.

Similarly $V(Y) = n\theta(1-\theta)$ is the Variance of a Binomial random Variable with Parameter n and θ.

Hence $Y \sim B(n, \theta)$.

Theorem

The probability density function (pdf) of the sum of two independent continuous random Variables X and Y is given by the convolution of the pdfs.

Convolution of Probability Mass functions

Let X be a discrete r.v. with pmf $P_X(x)$ and Y be another discrete r.v independent of X with pmf $P_Y(y)$. Then the pmf of $Z = X + Y$ can be derived by using one of the following.

$$P_Z(z) = \sum_{y \in R_Y} P_X(z-y) P_Y(y)$$

or

$$P_Z(z) = \sum_{x \in R_X} P_Y(z-x) P_X(x).$$

The above is called convolution.

If X and Y are continuous random variables and are independent with pdf $f_X(x)$ and $f_Y(y)$, respectively. The convolution formula for $Z = X + Y$ will be

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f(y)\, dy.$$

or

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z-x) f(x)\, dx$$

Example (discrete case)

Let X be a discrete variable with pmf

$$P_X(x) = \begin{cases} \frac{1}{2}, & x = 1, 2 \\ 0, & \text{otherwise} \end{cases}$$

Let Y be another discrete variable, independent of X with pmf

$$P_Y(y) = \begin{cases} \frac{1}{3}, & x = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$$

Then let $R_Z = \{0, 1, 2, 3, 4\}$

If $Z = X + Y$, find the pdf of Z.

Soln

Using the convolution theorem.

$$P_Z(z) = \sum_{R_Y} P_X(z-y) P_Y(y)$$

$$= \sum_{y=0}^{2} P_X(z-y) P_Y(y)$$

Thm. If $z \in R_Z$: for $z = 0$, we have

$$P_Z(0) = \sum_{y=0}^{1} P_X(0-y) P_Y(y)$$

$$= 0\, P_Y(0) + P_X(0-1) P_Y(y) + P_X(0-2) P_Y(2)$$

$$= (0 \cdot \tfrac{1}{3}) + P_X(-1) P_Y(y) + P_X(-2) P_Y(2)$$

$$= 0 + (0 \cdot \tfrac{1}{3}) + (0 \cdot 1/3) = 0$$

for $z = 1$, we have

$$P_Z(1) = \sum_{y=0}^{2} P_X(1-y) P_Y(y)$$

$$= P_X(1-0) P_Y(0) + P_X(1-1) P_Y(1) + P_X(1-2) P_Y(2)$$

$$= P_X(1) \cdot P_Y(0) + P_X(0) P_Y(1) + P_X(-1) P_Y(2)$$

$$= \left(\frac{1}{2} \cdot \frac{1}{3}\right) + \left(0 \cdot \frac{1}{3}\right) + \left(0 \cdot \frac{1}{3}\right)$$

$$= \frac{1}{6}$$

for $z = 2$, the pmf is

$$P_Z(2) = \sum_{y=0}^{2} P_X(2-y) P_Y(y)$$

$$= P_X(2-0) P_Y(0) + P_X(2-1) P_Y(1) + P_X(2-2) P_Y(2)$$

$$= P_X(2) \cdot P_Y(0) + P_X(1) \cdot P_Y(1) + P_X(0) \cdot P_Y(2)$$

$$= \left(\frac{1}{2} \cdot \frac{1}{3}\right) + \left(\frac{1}{2} \cdot \frac{1}{3}\right) + \left(0 \cdot \frac{1}{3}\right)$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

For $z = 3$, the pmf

$$P_Z(3) = \sum_{y=0}^{2} P_X(3-y) P_Y(y).$$

$$= P_X(3) \cdot P_Y(0) + P_X(2) \cdot P_Y(1) + P_X(1) \cdot P_Y(2)$$

$$= 0 + \left(\frac{1}{2} \cdot \frac{1}{3}\right) + \left(\frac{1}{2} \cdot \frac{1}{3}\right)$$

Finally, For $Z = 4$, the PMF of Z is

$$P_Z(4) = P_X(4) \cdot P_Y(0) + P_X(3) \cdot P_Y(1) + P_X(2) \cdot P_Y(2)$$

$$= 0 + 0 + \frac{1}{2} \cdot \frac{1}{3}$$

$$= \frac{1}{6}$$

Thus the PMF of Z

Z	0	1	2	3	4
$P_Z(z)$	0	1/6	1/3	1/3	1/6

Example (Continuous Case)

Let X be a continuous random Variable with pdf

$$f_X(x) = \begin{cases} 2e^{-2x}, & 0 \le x \le \infty \\ 0, & 0 \end{cases}$$

Suppose Y has a Uniform distribution and independent of X, $0 \le Y \le 1$.

Define $Z = X + Y$, with $R_Z = [0, \infty)$

Find the pdf of Z.

For $z \in R_z$, the pdf of Z is

$$f_Z(z) = \int_y f_X(z-y) f_Y(y)\, dy$$

$$= \int_0^1 f_X(z-y)\, dy$$

$$= \int_0^1 2e^{-2(z-y)} \mathbb{I}_{\{z-y \geq 0\}}\, dy$$

$$= 2\int_0^1 (e^{-2z} \cdot e^{2y}) \mathbb{I}_{\{y \leq z\}}\, dy$$

$$= 2e^{-2z} \int_0^{1} e^{2y} \mathbb{I}_{\{y \leq z\}}\, dy$$

$$= 2e^{-2z} \int_0^{\min(z,1)} e^{2y}\, dy$$

$$= 2e^{-2z} \left[\frac{1}{2} e^{2y}\right]_0^{\min(z,1)}$$

$$= 2e^{-2z} \left[\frac{1}{2} e^{2\min(z,1)} - \frac{1}{2}\right]$$

$$= e^{-2z} \left[e^{2\min(z,1)} - 1\right]$$

$\therefore$ The pdf of Z is

$$f_Z(z) = \begin{cases} \exp(-2z)\left[\exp(2\min(z,1)) - 1\right], & z \in R_z \\ 0, & \text{o.w.} \end{cases}$$

Example 2 (Continuous case)

Let X and Y be i.i.d. exponential RVs with pdf

$$f_X(x) = \begin{cases} e^{-x}, & 0 \le x < \infty \\ 0, & \text{otherwise} \end{cases}$$

and

$$f_Y(y) = \begin{cases} e^{-y}, & 0 \le y < \infty \\ 0, & \text{otherwise} \end{cases}$$

Define $Z = X + Y$, $0 \le z < \infty$.

Then for $z \in R_Z$, the pdf of Z is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy$$

$$= \int_{0}^{\infty} f_X(z-y) e^{-y}\, dy$$

$$= \int_{0}^{\infty} e^{-(z-y)} e^{-y}\, dy \; I_{[z-y \ge 0]}$$

$$= \int_{0}^{\infty} e^{(-z+y)} e^{-y}\, I_{[y \le z]}\, dy$$

$$= \int_{0}^{z} e^{(-z+y)} e^{-y}\, dy$$

$$= \int_{0}^{z} e^{-z}\, dy$$

$$= e^{-z} \int_{0}^{z} dy$$

$$= z e^{-z}, \quad z \in R_Z.$$